

Changwoo Lee

Curriculum Vitae

CONTACT INFORMATION

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RESEARCH INTERESTS

Efficient Large Language Model Inference, Structured Matrix, Hardware-aware LLM Inference Methods.

PROFESSIONAL EXPERIENCE

Machine Learning Researcher

Jun 2026 – Present

Qualcomm, San Diego CA USA

- Investigates foundations and techniques for efficient LLM inference.

Research Intern

May 2025 – Aug 2025

Google Deep Mind, Mountain View, CA, USA

- Studied a dynamic inference algorithm to improve the efficiency of audio encoders of multi-modal LLMs.

NLP Research Intern

May 2024 – Aug 2024

Samsung Research America, Mountain View, CA, USA

- Investigated efficient fine-tuning methods for on-device foundation models.

EDUCATION

University of Michigan, Ann Arbor, Michigan, USA

- Ph.D. in Electrical and Computer Engineering Aug 2020 – May 2026
 - Advisor: Prof. Hun-Seok Kim
 - Thesis Title: Learning Low-dimensional Structures for Efficient Deep Learning

PUBLICATIONS

- P. Abillama, **C. Lee**, J. Dong, D. Blaauw, D. Sylvester and H. Kim. “Accelerating Block Low-Rank Foundation Model Inference on Memory-Constrained GPUs.” In High-Performance Parallel and Distributed Computing (HPDC), 2026. [Paper]
- C. Yaras, A. S. Xu, P. Abillama, **C. Lee**, and L. Balzano. “MonarchAttention: Zero-Shot Conversion to Fast, Hardware-Aware Structured Attention.” *NeurIPS 2025* (Spotlight). [Paper]
- P. Abillama, **C. Lee**, A. Bejarano-Carbo, D. Blaauw, and H. Kim. “One-Hot Multi-Level Leaky Integrate-and-Fire Spiking Neural Networks for Enhanced Accuracy-Latency Tradeoff.” *IEEE Access*, 2025.
- **C. Lee**, S. Kwon, Q. Qu, and H. Kim. “BLAST: Block-Level Adaptive Structured Matrices for Efficient Deep Neural Network Inference.” *Neural Information Processing Systems (NeurIPS)*, 2024. [Paper]
- Z. Fan, H. An, B. Xu, L. Xu, C.W. Tseng, Y. Peng, A. Bejarano-Carbo, P. Abillama, A. Cao, B. Liu, **C. Lee**, Z. Wang, H. Kim, D. Blaauw, and D. Sylvester. “AIMMI: Audio and Image Multi-Modal Intelligence via a Low-Power SoC With 2-MByte On-Chip MRAM for IoT Devices.” *IEEE Journal of Solid-State Circuits (JSSC)*, 2024. [Paper]
- **C. Lee** and H. Kim. “Differentiable Learning of Generalized Structured Matrices for Efficient Deep Neural Networks.” In *International Conference on Learning Representations (ICLR)*, 2024. [Paper]
- Z. Fan, Q. Zhang, P. Abillama, S. Shoouri, **C. Lee**, D. Blaauw, H. Kim, and D. Sylvester. “TaskFusion: An Efficient Transfer Learning Architecture with Dual Delta Sparsity for Multi-Task Natural Language Processing.” In *Proceedings of the 50th Annual International Symposium on Computer Architecture (ISCA)*, 2023. [Paper]

- C. Bian, C. Hsu, **C. Lee**, and H. Kim. “Learning-Based Near-Orthogonal Superposition Code for MIMO Short Message Transmission.” In *IEEE Transactions on Communications (TCOM)*, 2023. [[Paper](#)]
- **C. Lee**, X. Hu, and H. Kim “Deep Joint Source-Channel Coding with Iterative Source Error Correction.” In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023. [[Paper](#)]
- Z. Fan, H. An, Q. Zhang, B. Xu, L. Xu, C. Tseng, Y. Peng, A. Cao, B. Liu, **C. Lee**, Z. Wang, F. Liu, G. Wang, S. Jiang, H. Kim, D. Blaauw, D. Sylvester. “Audio and Image Cross-Modal Intelligence via a 10TOPS/W 22nm SoC with Back-Propagation and Dynamic Power Gating.” In *IEEE Symposium on VLSI Circuits (VLSI-Symposium)*, 2022. [[Paper](#)]
- D. Kim, **C. Lee**, and K.S. Chung. “A Confidence-Calibrated MOBA Game Winner Predictor.” In *IEEE Conference on Games (CoG)*, 2020. [[Paper](#)]
- **C. Lee**, and K.S. Chung. “GRAM: Gradient Rescaling Attention Model for Data Uncertainty Estimation in Single Image Super Resolution.” In *IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2019. [[Paper](#)]

SKILLS **Programming Languages and Frameworks.** Python, PyTorch, NumPy, Hugging Face Transformers,
SoC Design, Hanyang University Spring 2018

SERVICE **Review Experience.**
 • NeurIPS 2026, ICML 2026, ICLR 2026, NeurIPS 2025, ICML 2024, AISTATS 2024, ICLR 2024, NeurIPS 2024, IEEE Transactions on Communications, IEEE Transactions on Mobile Computing